Right-Triangle Trigonometry

ANSWER KEY

Section 1: The Six Trigonometric Ratios

1 In a right triangle with acute angle θ, the side opposite θ is 5 and the hypotenuse is 13. Find $\cos\theta$.

a. 12/13 ✓

b. 5/12

c. 13/12

d. 12/5

Adjacent side $a = \sqrt{13^2 - 5^2} = \sqrt{169 - 25} = 12$. $\cos\theta = \frac{\text{adjacent}}{\text{hypotenuse}} = 12/13$.

2 For the same triangle ($extopp = 5$, $\text{adj} = 12$, $\text{hyp} = 13$), find $\sec\theta$.

a. 13/12 ✓

b. 13/5

c. 12/13

d. 12/5

$\sec\theta = \frac{1}{\cos\theta} = \frac{\text{hypotenuse}}{\text{adjacent}} = \frac{13}{12}$.

3 If $\tan\theta = 3/4$ in a right triangle, find $\csc\theta$.

a. 5/3 ✓

b. 5/4

c. 4/3

d. 3/5

Opposite $= 3$, adjacent $= 4 \Rightarrow$ hypotenuse $= 5$. $\csc\theta = \frac{\text{hypotenuse}}{\text{opposite}} = 5/3$.

Section 2: Special Right Triangles (30-60-90 and 45-45-90)

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4 In a $30^\circ - 60^\circ - 90^\circ$ triangle, if the shorter leg is 4, find the lengths of the longer leg and the hypotenuse.

a. **Longer leg** $= 4\sqrt{3}$**, Hypotenuse** $= 8$ ✓

b. Longer leg $= 8$, Hypotenuse $= 4\sqrt{3}$

c. Longer leg $= 4\sqrt{2}$, Hypotenuse $= 8$

d. Longer leg $= 6$, Hypotenuse $= 8$

Ratio of sides opposite $30^\circ : 60^\circ : 90^\circ$ is $1 : \sqrt{3} : 2$. With short leg 4: longer leg is $4\sqrt{3}$ and hypotenuse is 8.

5 In a $45^\circ - 45^\circ - 90^\circ$ triangle, if the hypotenuse is 10, find the length of each leg.

a. $5\sqrt{2}$ ✓

b. 5

c. $10\sqrt{2}$

d. $5\sqrt{3}$

Hypotenuse $= s\sqrt{2} = 10 \Rightarrow s = \frac{10}{\sqrt{2}} = 5\sqrt{2}$.

6 What is the exact value of $\sec(60^\circ) + \tan(45^\circ)$?

a. 3 ✓

b. $2 + \sqrt{3}$

c. $1 + \sqrt{3}$

d. 2

$\cos(60^\circ) = 1/2 \Rightarrow \sec(60^\circ) = 2$. $\tan(45^\circ) = 1$. Sum $= 2 + 1 = 3$.

Section 3: Cofunction Identities and Reciprocals

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7 Which cofunction identity correctly relates sine and cosine for complementary acute angles?

a. $\sin(90^\circ - \theta) = \cos\theta$ ✓

b. $\sin(90^\circ - \theta) = -\cos\theta$

c. $\sin(90^\circ - \theta) = \sin\theta$

d. $\sin(90^\circ - \theta) = \tan\theta$

In a right triangle, the two acute angles are complementary, so the side opposite one angle is adjacent to the other: $\sin(90^\circ - \theta) = \cos\theta$.

8 If $\cos(28^\circ) \approx 0.8829$, what is $\sin(62^\circ)$?

a. 0.8829 ✓

b. 0.1171

c. 0.4695

d. 1.1326

Since $28^\circ + 62^\circ = 90^\circ$, $\sin(62^\circ) = \cos(90^\circ - 62^\circ) = \cos(28^\circ) \approx 0.8829$.

9 Simplify $\cot(90^\circ - \theta) \cdot \cot\theta$.

a. 1 ✓

b. $\tan^2\theta$

c. $\cot^2\theta$

d. 0

By cofunction identity, $\cot(90^\circ - \theta) = \tan\theta$. Then $\tan\theta \cdot \cot\theta = \tan\theta \cdot \frac{1}{\tan\theta} = 1$.

Section 4: Angles of Elevation and Depression

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10 From a point 50 ft from the base of a building, the angle of elevation to the top is 60°. How tall is the building?

a. $50\sqrt{3}$ ft ✓

b. 100 ft

c. 50 ft

d. $25\sqrt{3}$ ft

$\tan(60^\circ) = \frac{h}{50} \Rightarrow h = 50\tan(60^\circ) = 50\sqrt{3}$ ft.

11 A lighthouse keeper 120 ft above sea level spots a boat at an angle of depression of 30°. How far is the boat from the base of the lighthouse?

a. $120\sqrt{3}$ ft ✓

b. $60\sqrt{3}$ ft

c. 240 ft

d. 120 ft

The angle of elevation from the boat equals the angle of depression: 30°.
$\tan(30^\circ) = \frac{120}{d} \Rightarrow d = \frac{120}{\tan(30^\circ)} = \frac{120}{1/\sqrt{3}} = 120\sqrt{3}$ ft.

12 A 20-ft ladder leans against a wall reaching a height of 16 ft. What is $\sin\theta$ where θ is the angle the ladder makes with the ground?

a. 4/5 ✓

b. 3/5

c. 4/3

d. 3/4

$\sin\theta = \frac{\text{height}}{\text{length}} = \frac{16}{20} = \frac{4}{5}$.

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